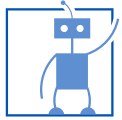


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Cognitive Systems: Neural Networks and Applied AI

Reinforcement Learning: Dynamic Programming, Policy Iteration, Value Iteration

Prof. Dr.-Ing. habil. Alois Knoll

Mohammadhossein Malmir, M.Sc.

Agenda

- Dynamic Programming (DP)
 - **DP Prediction: Policy Evaluation**
 - DP Control: Policy Improvement
 - DP Control: Policy Iteration
 - DP Control: Value Iteration

DP Prediction

- For a known MDP ($\langle S, A, P, R, \gamma \rangle$), we can simply evaluate the value function for an arbitrary policy by iteratively sweeping over all states:

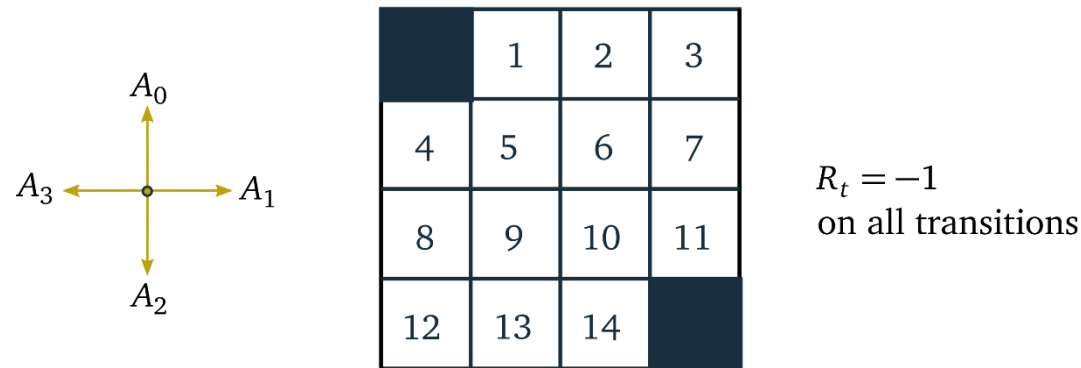
$$\begin{aligned}
 v_{k+1}(s) &\doteq \mathbb{E}_{\pi} [R_{t+1} + \gamma v_k(S_{t+1}) | S_t = s] \\
 &= \sum_a \pi(a|s) \sum_{s', r} p(s', r | s, a) [r + \gamma v_k(s')]
 \end{aligned}$$

- Propagating value between consecutive states by iteratively exploiting the recursive relationship that is formulated by the Bellman Equation is denoted as **bootstrapping**.

γ	discount rate	S	stochastic state function
a	particular action	s	particular state
R	stochastic reward function	v	value (expected return)
r	expected reward	π	policy

[§]Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, **p.74**.
Slide is adapted from CogSys21 RL lecture held by Prof. Dr.-Ing. Noah Klarmann.

DP Prediction: Gridworld Example



v_0	v_1	v_2	v_3	v_{10}	v_∞
0.0 0.0 0.0 0.0	0.0 -1.0 -1.0 -1.0	0.0 -1.7 -2.0 -2.0	0.0 -2.4 -2.9 -3.0	0.0 -6.1 -8.4 -9.0	0.0 -14. -20. -22.
0.0 0.0 0.0 0.0	-1.0 -1.0 -1.0 -1.0	-1.7 -2.0 -2.0 -2.0	-2.4 -2.9 -3.0 -2.9	-6.1 -7.7 -8.4 -8.4	-14. -18. -20. -20.
0.0 0.0 0.0 0.0	-1.0 -1.0 -1.0 -1.0	-2.0 -2.0 -2.0 -1.7	-2.9 -3.0 -2.9 -2.4	-8.4 -8.4 -7.7 -6.1	-20. -20. -18. -14.
0.0 0.0 0.0 0.0	-1.0 -1.0 -1.0 0.0	-2.0 -2.0 -1.7 0.0	-3.0 -2.9 -2.4 0.0	-9.0 -8.4 -6.1 0.0	-22. -20. -14. 0.0

A stochastic action function
 R stochastic reward function
 v value (expected return)

[§]Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, **p.76,77**.
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Agenda

- Dynamic Programming (DP)
 - ✓ DP Prediction: Policy Evaluation
 - **DP Control: Policy Improvement**
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 - DP Control: Value Iteration

DP Control: Policy Improvement

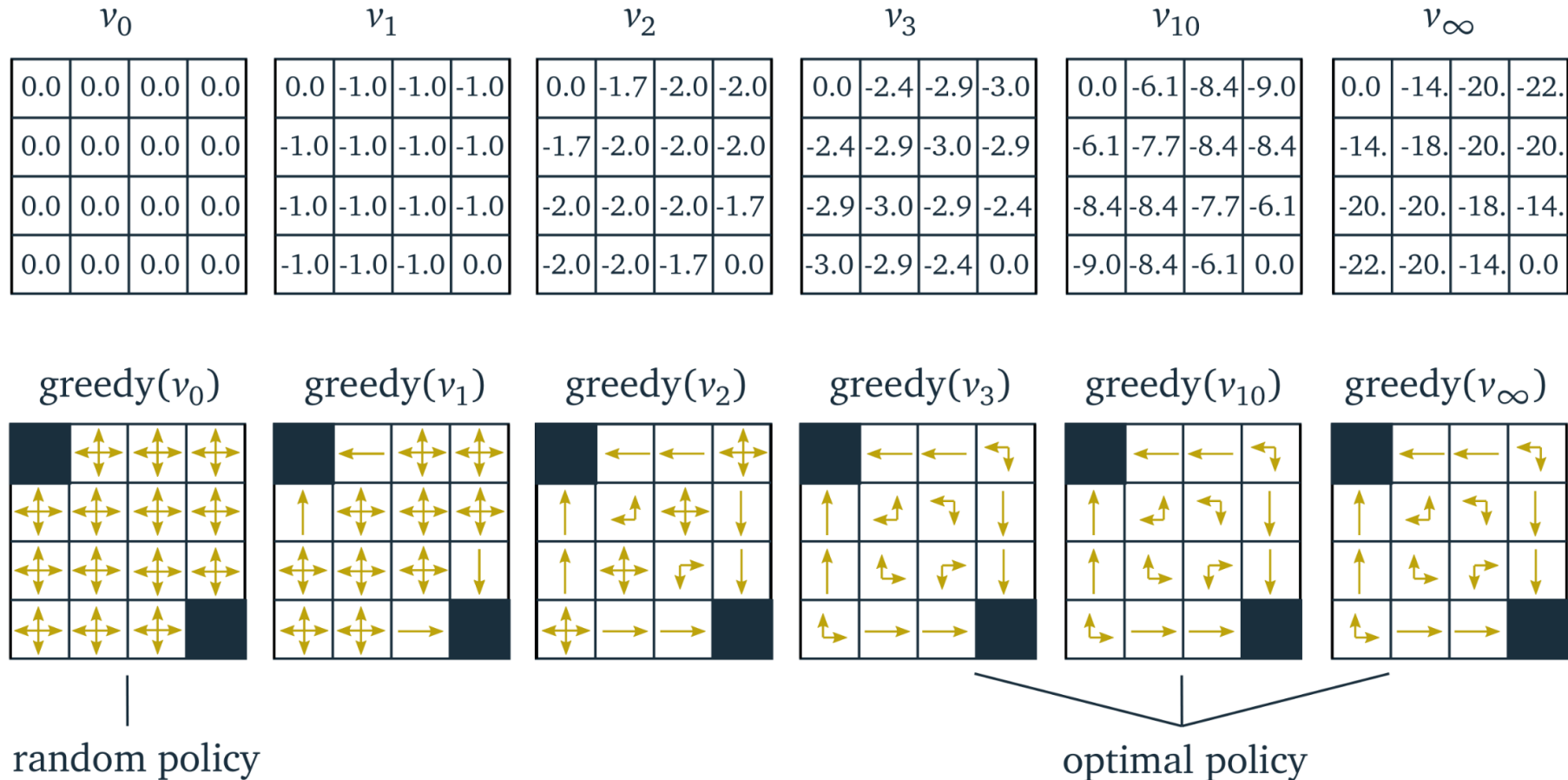
- We can simply improve our policy π by greedily acting on the value function v :

$$\begin{aligned}
 \pi'(s) &\doteq \arg \max_a q_{\pi}(s, a) \\
 &= \arg \max_a \mathbb{E} [R_{t+1} + \gamma v_{\pi}(S_{t+1}) | S_t = s, A_t = a] \\
 &= \arg \max_a \sum_{s', r} p(s', r | s, a) [r + \gamma v_{\pi}(s')]
 \end{aligned}$$

π	policy	R	stochastic reward function
γ	discount rate	r	expected reward
A	stochastic action function	S	stochastic state function
a	particular action	s	particular state
q	value for a specific action	v	value (expected return)

[§]Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, **p.79**.
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DP Control: Gridworld Example



v value (expected return)

^sSutton, R. S. and Barto, A. G. Reinforcement learning: An introduction. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, **p.76,77**.
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- Dynamic Programming (DP)
 - ✓ DP Prediction: Policy Evaluation
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DP Control: Policy Iteration

- **Policy iteration** is alternating between an **exhaustive evaluation** step and a greedy **policy improvement** step afterwards (GPI):

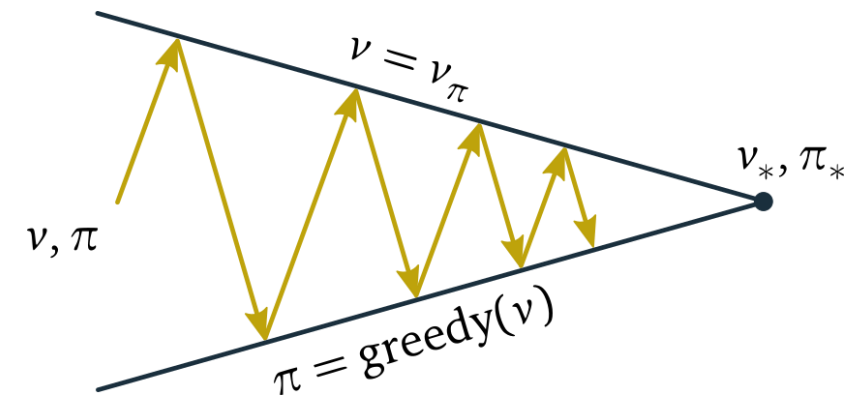
$$\pi_0 \xrightarrow{\text{eval.}} v_{\pi_0} \xrightarrow{\text{impr.}} \pi_1 \xrightarrow{\text{eval.}} v_{\pi_1} \xrightarrow{\text{impr.}} \pi_2 \dots \xrightarrow{\text{impr.}} \pi_* \xrightarrow{\text{eval.}} v_*$$

- evaluation: repeat value updates till $v_{k+1}(s) \approx v_k(s)$

$$v_{k+1}(s) = \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a) [r + \gamma v_k(s')]$$

- improvement: update to greedy policy at the current iteration

$$\pi'(s) = \arg \max_a \sum_{s',r} p(s',r|s,a) [r + \gamma v_{\pi}(s')]$$



[§]Sutton, R. S. and Barto, A. G. Reinforcement learning: An introduction. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, **p.80**.

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r	expected reward	v	value (expected return)
a	particular action	π	policy
s	particular state	γ	discount rate

Agenda

- Dynamic Programming (DP)
 - ✓ DP Prediction: Policy Evaluation
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 - **DP Control: Value Iteration**

DP Control: Value Iteration

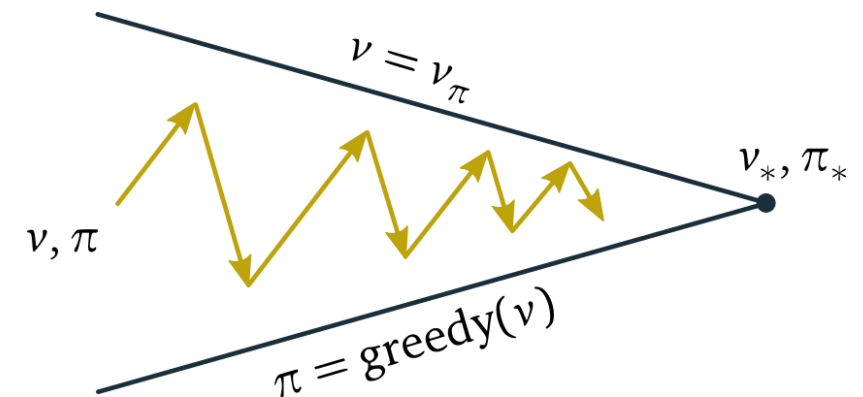
- A variation of policy iteration that is not using exhaustive evaluation step but a single sweep is called **value iteration** (converges to v_*).
- single update for combined policy evaluation and improvement step:

$$v_{k+1}(s) \doteq \max_a \mathbb{E} [R_{t+1} + \gamma v_k(S_{t+1}) | S_t = s, A_t = a]$$

$$= \max_a \sum_{s', r} p(s', r | s, a) [r + \gamma v_k(s')]$$

- after convergence ($v_{k+1}(s) \approx v_k(s)$):

$$\pi_*(s) = \arg \max_a \sum_{s', r} p(s', r | s, a) [r + \gamma v_*(s')]$$



[§]Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, **p.83**.

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A	stochastic action function	r	expected reward	v	value (expected return)
a	particular action	S	stochastic state function	π	policy
R	stochastic reward function	s	particular state	γ	discount rate